

Footprints and Handprints

How carbon and ecological accounting work,
and how to read the numbers they produce

Frederik Berg · July 2026

Written with AI research and drafting assistance; see the note on sources and method at the end

Summary

Ask how large one country's carbon footprint is per person and you get several different answers, all of them correct. For Germany the published figures run from 6.8 to 10.4 tonnes a year. They differ because they answer different questions: what is emitted inside the border, which gases are counted, whether goods bought from abroad are included, and whether each resident is charged a share of roads, hospitals and public administration. A footprint figure quoted without those four properties cannot be compared with anything.

Two measurements share the word "footprint" and are not versions of each other. The carbon footprint is a weight of greenhouse gases. The ecological footprint is an area of productive land and sea, and it converts carbon dioxide into the forest area that would be needed to absorb it. That conversion turns out to dominate: about 60 per cent of the ecological footprint is the carbon term, and the other five components have barely moved in sixty years. The metric works as a way to show that consumption exceeds what the planet regenerates. It does not work for choosing between two courses of action, because the assumption inside its largest component moves the answer more than the options being compared.

For an individual product, what gets counted depends on the method. Tracing a supply chain step by step is precise about what it includes and has to stop somewhere, typically before the factory building and the machines, which understates the total by twenty to fifty per cent. Working from the whole economy instead misses nothing, because every purchase a company ever made is recorded somewhere, but it treats everything within an industry as identical. Both are standard; they fail in opposite directions.

The handprint counts emissions avoided at other people rather than emissions caused. It exists because a personal footprint cannot show the largest thing an individual does — persuading a landlord, an employer or a council to change something structural. Unlike a footprint it is deliberately not exclusive: two people can claim the same avoided tonne, so the sum of everyone's handprints can exceed the emissions that exist. That makes it useful for describing influence and invalid to subtract from a footprint.

1 Two measurements, one word

The **carbon footprint** is a weight. It counts greenhouse gases and reports them in tonnes of carbon dioxide equivalent, written t CO₂e. Methane and nitrous oxide are converted into the amount of CO₂ that would cause the same warming over a hundred years, so everything lands in one unit. Climate targets are written in this unit, which is why it is the one that matters when a decision depends on the answer. The rules for measuring a product this way are set by ISO 14067,

a narrowing of the general life-cycle assessment standards to the single question of climate impact [32, 30, 31].

The **ecological footprint** is an area. It asks how much biologically productive land and sea a population needs — to grow its food, catch its fish, cut its timber, stand its buildings on, and absorb its carbon dioxide — and reports the answer in global hectares, a hectare of world-average productivity. That demand is compared against *biocapacity*, the productive surface actually available. The accounts, developed by Mathis Wackernagel and William Rees and now maintained by the Global Footprint Network with York University, track six categories: cropland, grazing land, forest products, fishing grounds, built-up land, and the forest area needed to absorb emissions [12].

Neither number contains the other, and neither converts into the other. The ecological footprint does have a carbon term inside it, but that term is not the emissions — it is a hypothetical forest that would have to exist to absorb them. The carbon footprint, in turn, says nothing about cropland or fisheries. The two figures for one country in one year can move in opposite directions.

A carbon footprint asks how much something warmed the planet. An ecological footprint asks how much of the planet's productive surface it occupied. Only the first is denominated in the quantity climate targets are set in.

2 Why one country has four different numbers

Germany is a useful case because all four variants are published, and the gap between them is larger than most of the differences people argue about.

The narrowest counts only what is physically emitted inside the border, and only carbon dioxide: 6.77 tonnes per person in 2024 [7]. Add the other greenhouse gases — agricultural methane, nitrous oxide from fertiliser, refrigerant gases — and the same territorial basis gives 649 million tonnes for 2025, which across 83.5 million residents is 7.8 tonnes each [2, 6]. This is the number national climate law is written against; Germany's legal targets are at least 65 per cent below 1990 by 2030 and net zero by 2045 [3].

Now correct for trade. Subtract the emissions caused by things Germany makes and sells abroad, add the emissions caused abroad by things Germany buys, and the figure rises to 9.09 tonnes of CO₂ per person for 2023, against 7.02 tonnes territorial in the same year [7, 8]. Roughly three tonnes of every ten a German resident is responsible for are released outside Germany.

Finally the figure consumer calculators show: 10.4 tonnes. It is trade-corrected, counts all greenhouse gases, adds the extra warming aircraft cause at altitude, and gives every resident an equal share of emissions from public administration, health, education, water, waste and civil engineering [1, 5].

2.1 A share is not a footprint

Dividing a country's total emissions by its population produces a share, and a share is a different object from a footprint. It includes emissions from factories producing goods for export that no resident ever touches, and excludes emissions from a factory abroad producing something a resident bought last week.

It also hands everyone the same value, which is wrong in a way worth quantifying: in Germany the lowest-emitting tenth of the population sits near 7 tonnes and the highest-emitting tenth near 17.7, more than half again above the national average [4].

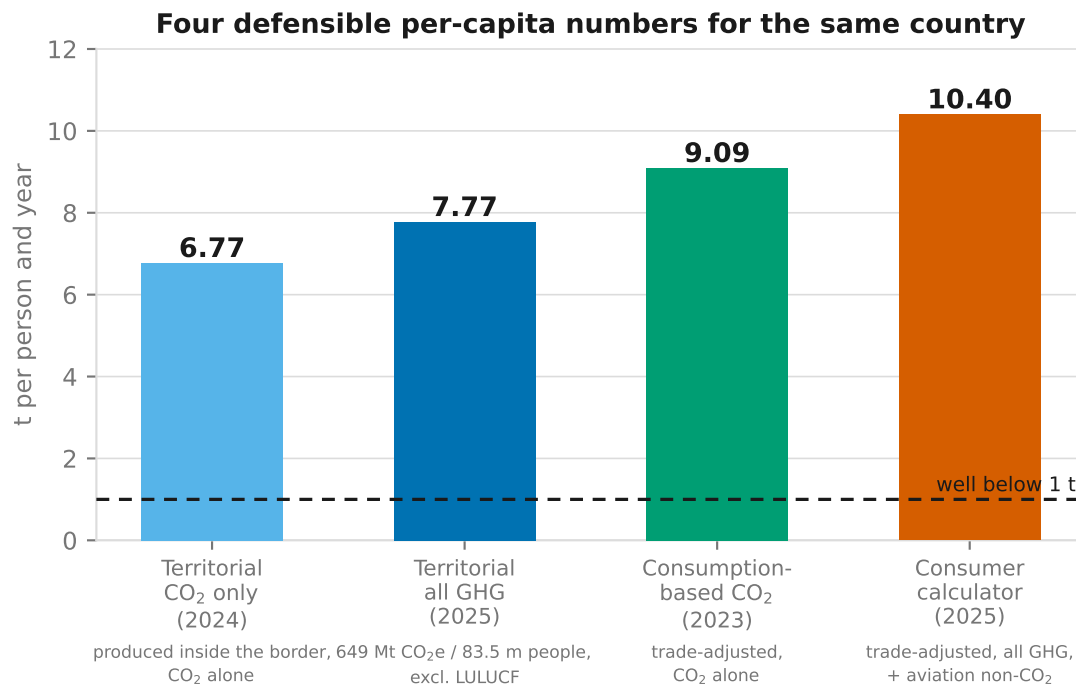


Figure 1: The same country, four accounting rules, no measurement error. The note under each bar states what that rule includes.

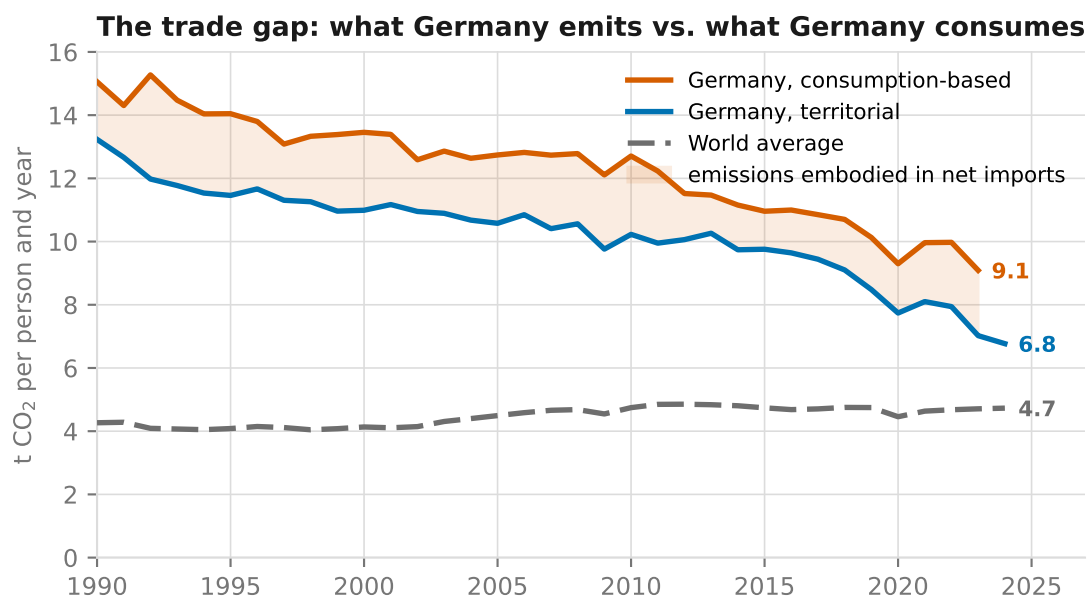


Figure 2: Germany's emissions inside its border have fallen faster than the emissions caused by German consumption. The shaded band is the part released abroad. Data: Our World in Data and the Global Carbon Project [7, 8].

3 The same question, worldwide

Global fossil carbon dioxide emissions reached a record 38.1 billion tonnes in 2025, with a further 4.1 billion from deforestation and other land-use change [10, 11]. Divided by population that is 4.73 tonnes of CO₂ per person, or 6.67 tonnes once all greenhouse gases are counted [7, 9].

The spread between countries is about a factor of twenty. Qatar emits 41.3 tonnes per person, the United States 14.2, China 8.7, Germany 6.8, France 4.0 and India 2.2 [7]. France sits at little more than half the German figure while being a comparably wealthy country, almost entirely because

of how it generates electricity — a reminder of how much of a national footprint is decided by infrastructure rather than by behaviour.

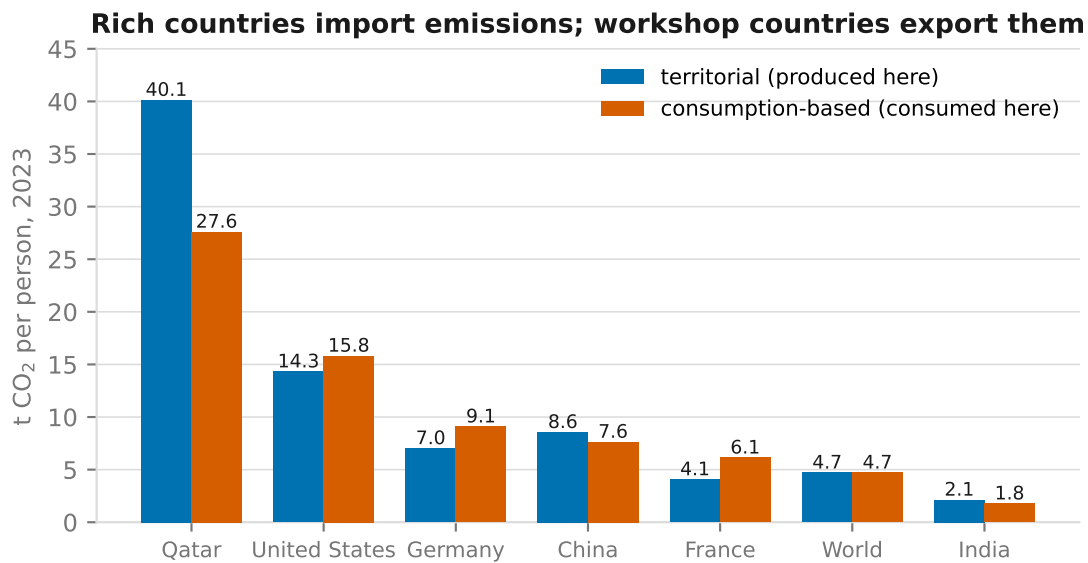


Figure 3: Emissions produced inside a country against emissions caused by its consumption, 2023. Rich countries import more embodied emissions than they export; manufacturing economies do the reverse. Data: [7, 8].

Country averages, though, hide a spread that is now larger than the spread between countries. In 2019 the poorest half of humanity accounted for 12 per cent of global emissions and the richest tenth for 48 per cent. Since 1990 the poorest half has been responsible for 16 per cent of the growth in emissions and the richest one per cent for 23 per cent. Most striking for how footprints are usually measured: the bulk of the top percentile's emissions comes from what they own and invest in rather than from what they consume [20].

Income, not nationality, is the strongest predictor of a footprint

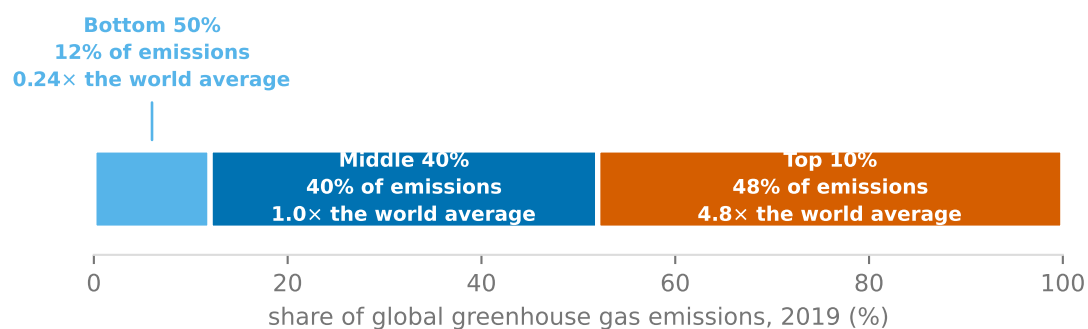


Figure 4: Shares of global greenhouse gas emissions by global income group, 2019. The multiplier is each group's share of emissions divided by its share of people. Data: Chancel (2022) [20].

That last point has a direct consequence. A calculator that asks about flights, heating and diet measures consumption. For most people consumption is the dominant term and the answer is close to right. For the top of the distribution it is not, and the group with the largest footprint is the one such a calculator understates most.

4 What sits inside a personal footprint

The 10.4-tonne figure is the trade-corrected one — the fourth bar in Figure 1, the number a consumer calculator reports. It is worth being explicit about how it is built, because two quite different procedures are at work inside it and neither is the one people usually assume.

For the parts a person can report about themselves, the calculation runs **bottom-up**: kilometres driven, litres of heating oil, kilowatt-hours of electricity, flights taken, each multiplied by an emission factor. For the parts nobody can report — the emissions behind a shirt, a restaurant meal, a hospital — it runs **top-down**: national totals for that sector, divided by the population, then scaled up or down by what the person says about their habits. Food, general consumption and public infrastructure are computed this way, as are the base loads inside housing and mobility [5]. Section 5 explains where those national totals come from.

So the answer to “is this per-product or is it the country divided by people?” is: both, in different categories. Nobody adds up the individual products they bought, because the data to do that does not exist. What varies with a person’s answers is how far above or below the national average each category lands.

The six categories the calculator uses are where most of the interpretation happens, and few of them are where intuition puts them.

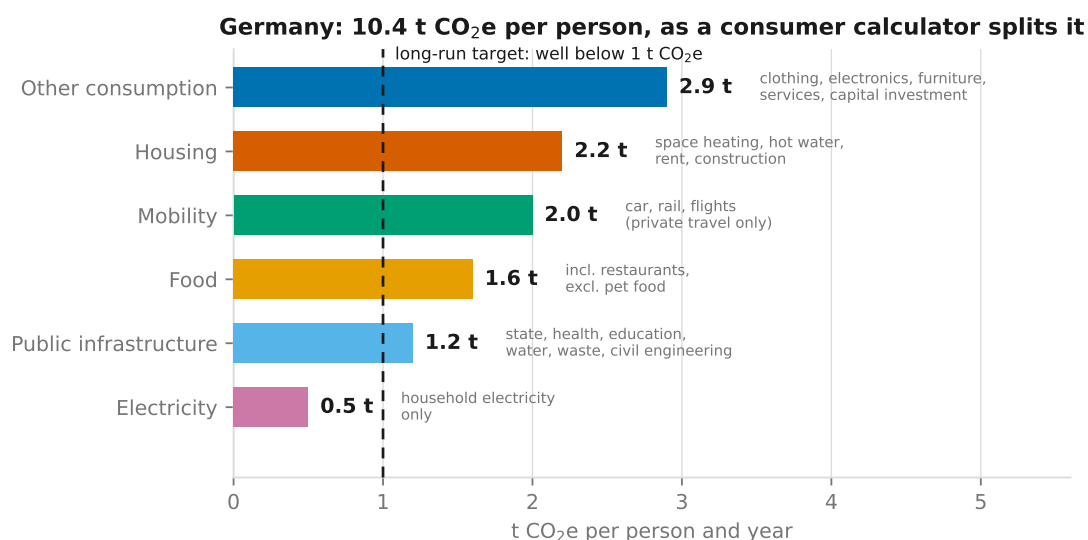


Figure 5: The average German footprint by category. The notes state what each one contains. Sources: [1, 5].

The largest single category is not heating and not driving. It is everything else bought — clothes, electronics, furniture, services — together with the machines and factory buildings that produced all of it. Housing comes second and means heating and hot water, plus renting and renovation work. Household electricity is a separate and surprisingly small category at half a tonne, which is the honest reason a green tariff moves a personal total less than people expect. Published savings figures for switching to renewable electricity are larger than this entire category, for a reason Section 7.1 takes up.

Mobility counts private travel only. A train taken to a client meeting is not in it; that emission is treated as business activity and reappears inside the consumption category. Food includes a proportional share of restaurants and canteens and excludes pet food, which also lands in consumption [5].

The last category is the one no user input can change. Public infrastructure, 1.2 tonnes, is an equal

share of the state itself: administration, health, education, water supply, waste disposal, and the construction of roads, railways, bridges and tunnels [5]. It sits underneath every resident regardless of how they live, and it is larger than the entire long-run target discussed in Section 7.

5 How a footprint is actually calculated

Two methods produce these numbers, and knowing which one produced a figure explains most of the disagreements between published sources.

5.1 Following the supply chain, and where it stops

The first method traces a product backwards step by step: cotton, spinning, dyeing, sewing, shipping. It is precise about what it includes, and it has to stop somewhere. Raw materials, process energy, direct emissions, packaging and transport are normally inside. The factory building, the machines, the administrative overhead and staff commuting are normally outside — commuting because it belongs to the organisation rather than to the product, the rest because allocating a thirty-year building across millions of units requires assumptions that are hard to defend.

The cost of stopping is measurable. Comparing traced inventories against whole-economy ones showed that this method systematically underestimates, in some product systems by half or more [37], and a group of the field's practitioners argued that where a study draws its boundary is not decided on scientific grounds at all [38]. The commonly cited range is a 20 to 50 per cent shortfall, and it is a shortfall rather than noise: the error only ever points one way.

Standards handle this differently, which is why two figures for one product can legitimately disagree. PAS 2050 leaves capital goods out by default [33]; the European Product Environmental Footprint method requires them [34]. For company reporting the Greenhouse Gas Protocol requires them and is strict about when: “companies should not depreciate, discount, or amortize the emissions from the production of capital goods over time” — a machine's full manufacturing emissions are booked in the year it is bought, not spread across its working life [35].

5.2 Working from the whole economy instead

The second method starts from national accounts — the tables recording which industries buy what from which other industries — and traces emissions through all of them. Nothing is cut off, because every purchase any company ever made is recorded somewhere. This is what stands behind the German calculator, which builds its sector averages from a model of 182 product groups [5].

The published worked example shows what that captures. German demand for research and development in 2020 came to 110.5 billion euros and carried 8.75 million tonnes of CO₂e, an intensity of 79 kilograms per thousand euros spent. Conferences those institutions hosted and business trips their staff took are inside that figure, as inputs bought from the hospitality and transport sectors [5]. Nothing has to be added by hand; the supply chain is already in the matrix.

The price of that completeness is bluntness. The same report notes that research needing a laboratory and research needing a desk carry the same coefficient [5]. Within a category everything is treated as identical.

Three further properties of this particular model bound what its numbers can support. Imported goods are assumed to have been made as cleanly as domestic ones, which understates exactly those sectors that have moved abroad. Emissions from land use and deforestation are excluded — as are, in exchange, the sinks from reforestation. And the data runs two to three years behind the year on

the label [5].

5.3 The factory question

Whether the factory that made a product is counted therefore has a specific answer that depends on the method, and in the whole-economy method the answer is yes with a qualification.

German investment in machines, equipment, vehicles and non-residential buildings came to roughly 153 million tonnes of CO₂e in 2021. Civil engineering is assigned to public infrastructure, residential construction to housing, and the majority stays inside general consumption on the reasoning that machinery and industrial buildings exist in order to produce consumer goods. Splitting it further onto individual products is stated to be impossible with the data that exists [5].

So the factory is in the number, but it is spread across the whole shopping basket rather than attached to the item bought.

5.4 Green electricity

The claim that renewable electricity makes an electricity footprint zero is a boundary question rather than a factual one. Company reporting rules require both figures to be published: one using the average emissions of the physical grid, and one derived from what was actually contracted, such as a supplier tariff or a renewable certificate [36].

The contracted figure can legitimately be close to zero. What it does not establish is that the purchase caused additional renewable generation to exist — the rules allocate existing generation to whoever contracted it and do not require the certificate to be additional. Two customers can therefore both lower their reported figure while total grid emissions are unchanged.

The German calculator handles the closest analogous case more conservatively. Electricity a household generates and feeds into the grid is credited as an avoidance *at other people*, calculated against the fossil generation it displaces, and is reported on a separate line rather than subtracted from the household's own footprint [5].

6 The ecological footprint, and what it is made of

Each component of the ecological footprint is consumption divided by a world-average yield, converted into global hectares. Biocapacity is the matching supply figure: area, adjusted for how productive that country's land of that type is [12].

The carbon component works differently from the other five. Emissions are reduced by the share the oceans absorb, and what remains is divided by the rate at which world-average forest absorbs carbon, producing an area of forest that would be needed but does not exist.

That component dominates the result. For Germany in 2023 — the last measured year in the current accounts — the footprint of consumption was 4.03 global hectares per person against a domestic biocapacity of 1.59. Carbon accounted for 2.43 of those 4.03, about 60 per cent. Cropland contributed 0.79, forest products 0.39, built-up land 0.19, grazing land 0.17 and fishing grounds 0.05 [12].

Remove the carbon layer and what remains — 1.60 global hectares — is close to what Germany's own land and sea regenerate. The five non-carbon components together have barely moved in six decades.

Worldwide the same accounts give 2.51 global hectares of demand per person for 2025 against 1.48

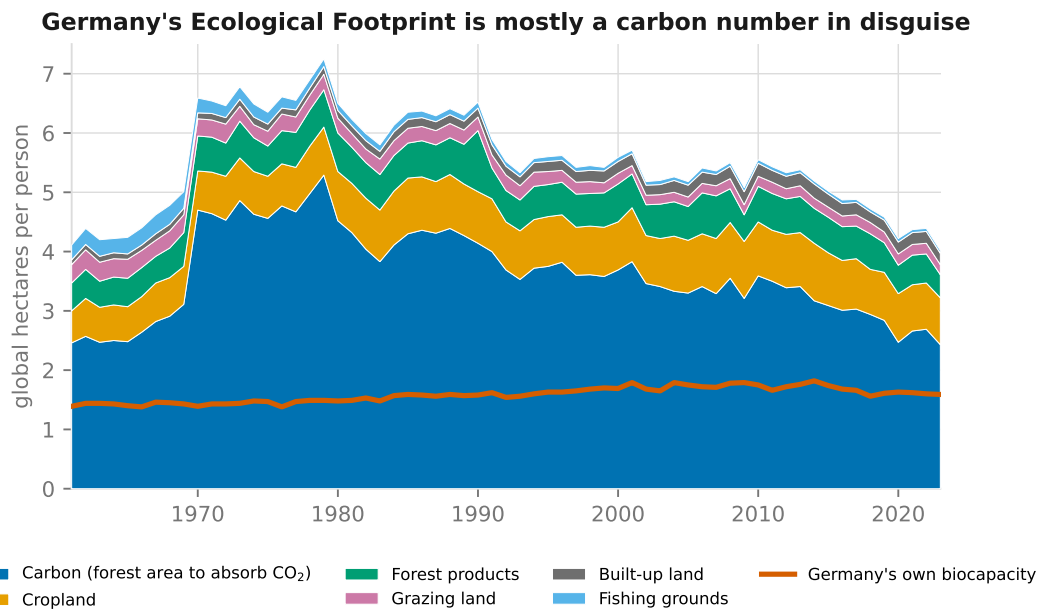


Figure 6: Germany's ecological footprint by component since 1961, against its own biocapacity. Almost all of the movement is the carbon layer at the bottom. Data: Global Footprint Network and York University [12].

available, a ratio of 1.70 that is the origin of the “1.7 Earths” figure. Demand first exceeded supply around 1970 and has not returned since [12].

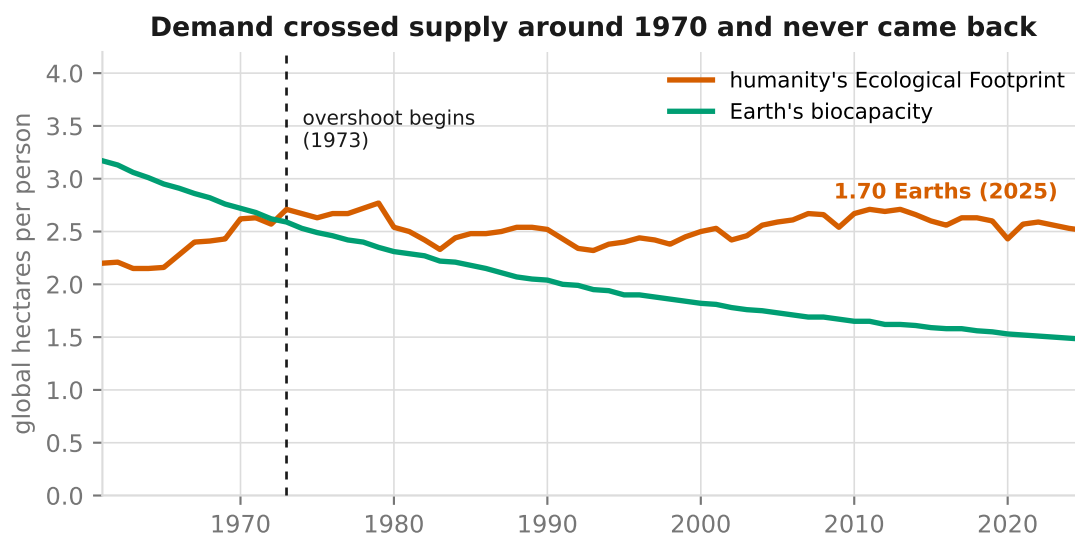


Figure 7: World ecological footprint and biocapacity per person. Biocapacity falls steadily because population grows faster than productive area. Data: [12].

Earth Overshoot Day — the date on which cumulative demand equals the year's biocapacity — falls on 30 July 2026 [13]. Germany's own date, meaning the date the world would reach if everyone consumed like a German resident, is 10 May [14].

Two properties of these dates are easy to misread. The whole series is recalculated every year, so a date published in an earlier year will not match the current one. When the 2026 accounts revised the ocean's carbon uptake upward, the date moved eight days later, while genuine deterioration moved it two days earlier — a net six days later than the previous estimate, in a year the accounts themselves describe as the highest overshoot ever recorded [13]. A later date can mean a better model rather than a better world. For the same reason, per-person hectare figures quoted from

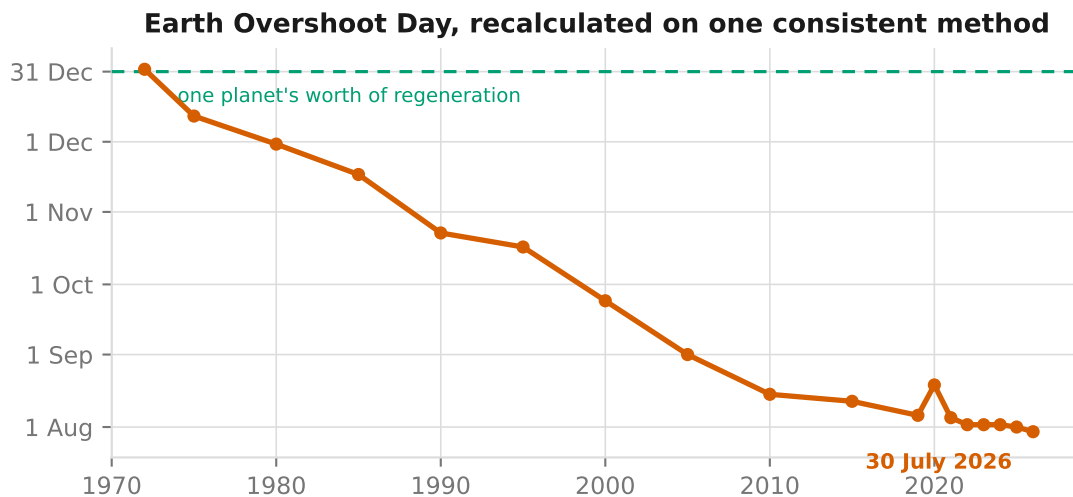


Figure 8: Earth Overshoot Day since 1972, recalculated on one consistent method. Data: [13].

different editions will not agree, and a comparison across sources has to check which edition each number came from.

6.1 The case against the metric, and what survives it

The ecological footprint has been criticised in the peer-reviewed literature for two decades, and the central criticism is arithmetic rather than rhetorical.

The headline claim is narrower than it sounds. “Humanity uses 1.7 Earths” invites the reading that we are consuming farmland, forests and fisheries at 1.7 times the rate they regrow. Break the number apart and that is not what it says. Blomqvist and colleagues showed that none of the five non-carbon categories runs a substantial deficit — cropland demand roughly matches cropland supply, fishing roughly matches fish, and so on. Essentially all of the measured overshoot is the carbon component [15]. The real claim is “we emit more CO₂ than forests absorb”, which is true, important, and already measurable in tonnes.

The hectares in that component do not exist, and how many there are is a choice. The carbon term is the forest area that *would be* needed to absorb the emissions. No such forest is planted; it is a unit conversion. How large it comes out depends on an assumed rate at which a hectare of forest takes up carbon, and on how much the oceans are assumed to absorb first. Neither is a measurement of what happened. Giampietro and Saltelli made this the core of their objection: the result swings with assumptions that cannot be pinned down, while the supply figure it is compared against is built with the same machinery [18]. The 2026 revision described above is the effect in practice — an updated ocean figure moved Earth Overshoot Day more than reality did.

The conversion licenses a conclusion nobody intends. Because emissions enter as “forest area required”, the arithmetic says global overshoot could be closed by planting enough fast-growing trees, with emissions unchanged. The atmosphere does not work that way; it responds to tonnes. A metric that can be satisfied without the physical problem improving is doing something odd with its unit.

Adding unlike things together loses the information. Cropland, fish and carbon are summed into one figure in hectares, so a rise in one can mask a fall in another and the total points at no particular action. Van den Bergh and Grazi argued on this basis that the aggregate offers nothing usable for policy [17].

Blomqvist and colleagues concluded that the metric as constructed is “so misleading as to preclude

their use in any serious science or policy context” [15].

Rees and Wackernagel replied that the metric answers one specific question — how much biocapacity does a population use against how much is available — and is being judged against a standard it never claimed to meet. The carbon share, they argued, reflects the real dominance of fossil fuels rather than a modelling artefact [16]. The Global Footprint Network’s own limitations note concedes that the metric does not measure pollution, biodiversity or water, and works best inside a set of indicators rather than alone [19].

Both positions are correct about different things, which points to a division of labour rather than a winner. To show a general audience in one number that consumption exceeds what the planet regenerates, the area framing and Overshoot Day do work that a tonnage cannot. To choose between two courses of action, the conversion to area adds an assumption without adding information, and the sensitivity of that assumption is larger than the differences being weighed.

7 What one person can change, and what they cannot

Two systematic reviews have measured the size of individual actions. A synthesis of 53 studies covering 771 options found the largest ones in transport — living without a car, switching to an electric vehicle, dropping one long return flight, each above 1.7 tonnes a year — followed by renewable electricity at 1.6 tonnes, home refurbishment at 0.9 and a vegan diet at 0.8 [22]. An earlier review of 148 scenarios put living car-free at 2.4 tonnes a year, avoiding one transatlantic return flight at 1.6 and a plant-based diet at 0.8 — four times more than comprehensive recycling and eight times more than changing lightbulbs [21].

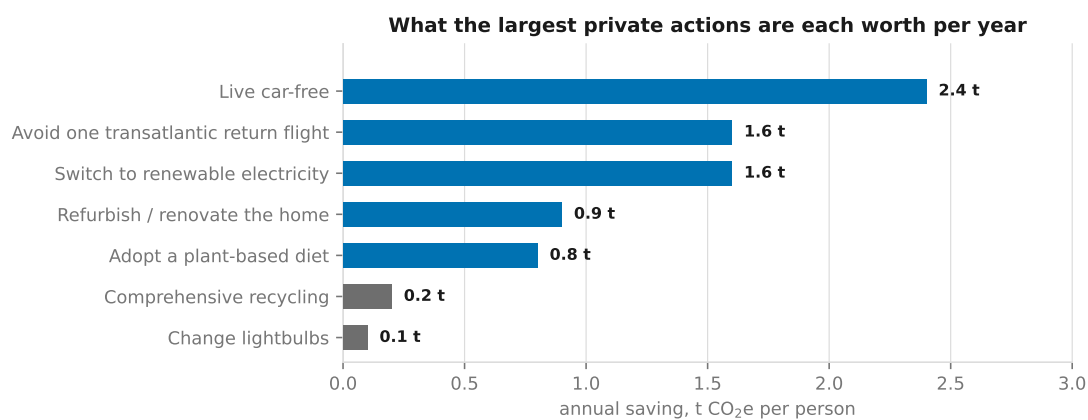


Figure 9: What each of the largest private actions is worth per year. Grey marks the two actions public campaigns push hardest. Sources: [22, 21].

7.1 The ranking is a population ranking, not a personal one

Every figure in that chart is a median across studies, measured against an average-emitting baseline. That makes the ordering correct for a population and unreliable for any particular person, because **the saving an action offers is capped by what that person already emits in the category**. Living car-free is worth 2.4 tonnes to someone who drives an average distance and nothing at all to someone who has no car. Ivanova and co-authors treat this explicitly, modelling geographical, technical and socio-demographic context as a determinant of how much any option delivers [22].

The chart in front of this section contains its own example. Switching to renewable electricity appears there at 1.6 tonnes — a median across studies from countries where household electricity is a much larger share of the footprint than it is in Germany, and where in several cases electricity

also supplies the heating. A German resident's entire household electricity category is 0.5 tonnes (Section 4). A saving cannot exceed the category it comes out of, so 0.5 is the ceiling on what that switch can deliver there, whatever the median says.

The effect is large enough to reorder the list. For a person without a car the chart's biggest item is worth nothing at all, which moves it from first place to last. Diet shows the same dependence in a milder form: measured against one consistent set of diets, the move from a medium meat-eater to a vegan is worth about 1.7 tonnes a year, while the move from a vegetarian to a vegan is worth about 0.6 — 7.04, 4.16 and 2.47 kg CO₂e per 2000 kcal respectively, annualised [23]. Same action, two starting points, a factor of nearly three between them.

For a German reader the diet step can be sized directly, and against something people have an intuition for. The calculator puts the average German food footprint at 1.6 tonnes a year (Section 4), and the environment agency estimates that moving from the average German diet to a vegan one cuts diet-related emissions by 38 to 52 per cent [25]. Combining the two gives roughly 0.6 to 0.8 tonnes a year. A return flight inside Europe of about 1000 kilometres each way — Frankfurt to Barcelona, Berlin to Rome — is some 2000 passenger-kilometres, and German aviation data puts short-haul economy travel at 195 grams of CO₂e per passenger-kilometre once non-CO₂ effects and fuel supply are counted [26], so about 0.4 tonnes. Giving up animal products for a year is therefore worth somewhere between one and a half and two such flights. Neither side of that comparison includes emissions from land-use change, which the calculator omits throughout and which is the larger omission on the food side [5].

So the published ordering describes a population, and which lever is largest for a given person depends on where that person already stands. A generic ranking names the categories that are usually large; only a personal baseline shows which of them still has room left in it.

7.2 Why the widely quoted child figure is missing from that chart

The same review lists a fifth action far above the others — having one fewer child, at 58.6 tonnes a year [21]. It is absent from the chart because it is a different kind of quantity from the rest, not simply a larger one.

The study underneath it computes a lifetime carbon legacy: under current conditions in the United States, each child adds about 9441 tonnes of CO₂ to an average woman's legacy, which is 5.7 times her own lifetime emissions [24]. Of that, half of a child's emissions are assigned to each parent, a quarter of the grandchildren's, and so on down the family tree [21]. Three things follow. It is a lifetime sum presented on an annual axis, where every other figure here is a saving that recurs each year. The child is charged its own emissions again once grown, so the same tonne sits in two ledgers, where every other figure here belongs to one person only. And it holds emission rates at today's level for as long as the family tree runs, which is the opposite of what the targets the literature exists to serve assume. None of this touches the four findings above, which are ordinary annual savings measured the ordinary way.

Adding the realistic levers together, allowing for the overlap between them and for the ceilings just described, gets somewhere around five to seven tonnes off a 10.4-tonne starting point. The total is indicative rather than exact — it is assembled from cross-country medians, and the German ceiling on one of its terms alone moves it by a tonne. The conclusion survives that imprecision comfortably: it is a substantial change in how a person lives, and it still does not reach the target.

The target itself is not a single published constant. Germany's environment agency states that no exact value can be derived, because it depends on population growth and on how many carbon sinks can be built, but that it must lie well below one tonne per person per year for net zero to

work — and that reaching it requires cutting more than 90 per cent from today's level [4]. An independent check gives the same order of magnitude: 170 billion tonnes of CO₂ remain in the budget for 1.5 degrees [10], which across roughly 8.2 billion people is about 21 tonnes each in total — under five years at the current German rate.

What is left after the private levers are exhausted is the electricity grid, the building stock, the railways, the freight system and industrial process heat. None of those is a consumption decision. The 1.2-tonne public share is not reachable by any of them either.

8 The handprint

The handprint answers the question that follows directly: if most of the remaining gap is structural, what is one person supposed to do about it?

It counts what someone changes at other people. Persuading a landlord to insulate a building cuts emissions for every household in it. Getting an employer to put solar on the roof, or a council to change a bus route, moves far more than any private decision. None of it appears in a personal footprint, which by construction only counts what passes through one person's own life. The concept originated at the Centre for Environment Education in India and was developed in Germany by Germanwatch with Brot für die Welt, who frame it as helping people “shape social and political change in their environment and thus change structural conditions in the long term” [27, 28]. Its corporate equivalent, avoided emissions, is defined as the difference a low-carbon product makes against a scenario without it, and is explicitly positioned as sitting outside a company's own emissions inventory [29].

The property that makes the concept work is also the one that limits it, and it is easiest to see with numbers.

Take a block of ten flats. One tenant spends a year pushing the landlord into insulating it, and afterwards the building burns half a tonne less CO₂e per flat per year. Five tonnes are genuinely avoided.

On the footprint side the bookkeeping is clean. Each of the ten residents now heats with less gas, so each of their personal footprints falls by 0.5 tonnes. Ten times 0.5 is 5, which is exactly what was avoided. Nothing is missing and nothing is counted twice — that is what a footprint is: every real tonne assigned to exactly one person, totals matching reality.

On the handprint side the tenant who organised it claims 5 tonnes, and that claim is honest too. Without them the insulation does not happen and none of the five tonnes is avoided.

Now add the two together and you get 10 tonnes from an event that saved 5. The tenant's own footprint also fell by 0.5, and that half-tonne now appears both in their footprint and inside their handprint. None of this is an error. The two numbers answer different questions — *how much do I cause?* and *how much did I change?* — and the second question cannot be answered without letting several people take credit for the same tonne.

The footprint concept assigns all of a population's emissions to individuals without overlap. The handprint cannot: avoidance achieved at other people cannot be assigned unambiguously to one actor and may be claimed by several at once, so the sum of all handprints can exceed the emissions that actually exist. Double counting is excluded from the footprint by construction and is inherent to the handprint — which is not a problem as long as the two are reported separately [5].

The rule that follows is hard: a handprint may be reported beside a footprint and never subtracted from it. The German calculator is built accordingly and quantifies only three cases — electricity fed into the grid, donations to climate projects, and climate-directed investments. A compensated flight still appears at full weight under mobility, with the compensation shown separately and negatively [5].

Once that split is clear, a large class of public claims sorts itself out. “Our product saved four million tonnes” is a handprint claim and says nothing about what the company emits. “We cut our emissions by four million tonnes” is a footprint claim and can be checked against a balance that has to add up. Claims that slide between the two are the standard shape of greenwashing, which is why the corporate guidance treats separate reporting as a condition of credibility [29].

9 Where the idea came from

A recurring argument holds that the personal carbon footprint is a public relations construct and should be distrusted on that basis. The documented facts support part of it. BP hired the advertising agency Ogilvy & Mather, ran an award-winning campaign from 2000, and launched a personal carbon footprint calculator in 2004 aimed at showing individuals how their daily life drives warming [39].

Two corrections matter. The term predates the campaign: the earliest use recorded by the Oxford English Dictionary is a 1999 issue of a BBC vegetarian cooking magazine, and the term became Oxford’s word of the year in 2007 [40]. The accounting predates it by longer — the ecological footprint dates from the early 1990s [12], and trade-corrected emission accounting is a standard application of input-output analysis, a technique from national statistics.

Where a word came from is not evidence about whether a measurement works. A statistical office computing consumption-based emissions performs the same arithmetic regardless of who once found the vocabulary convenient. What the objection does get right is the framing effect: a personal footprint presented without context invites the conclusion that the sum of private choices is the solution, and Section 7 puts that at roughly half. The response that follows is to keep the measurement and fix the framing, which is what the handprint was constructed to do.

10 Which metric answers which question

11 Conclusion

A footprint figure without its accounting rule is not information. The four German numbers span 6.8 to 10.4 tonnes and every one of them is correct. Before comparing any two figures, three properties have to be fixed: which gases, which boundary, and whether trade is counted. That single habit removes most of the confusion in public reporting, and it applies equally to hectare figures quoted from different editions of the ecological footprint accounts.

The metric should follow the question rather than the audience. Where a decision depends on the answer, tonnes of CO₂e with a stated boundary is the defensible unit. The ecological footprint earns its place as a way of making overshoot legible and loses it as soon as it is asked to rank options, for the reason its critics identified and its own accounts confirm: about 60 per cent of it is a carbon number in an area unit, and the rest has barely moved in sixty years.

The footprint and the handprint are complements, and the property that makes the handprint useful is the same one that makes it unsummable. Reported side by side they describe both what a person consumes and what a person changes. Netted against each other they describe nothing.

Table 1: Every row is a legitimate metric. The failure mode is using one to answer another's question.

Question	Metric	Why, and what it cannot do
Is a country meeting its legal target?	Territorial, t CO ₂ e	The basis of national inventories and climate law. Ignores trade entirely.
How much warming does a population's consumption cause?	Consumption-based, t CO ₂ e	Nets out exports, adds imports. Runs two to three years behind and inherits the bluntness of whole-economy methods.
Which of two products is better?	Product footprint, t CO ₂ e	Comparable only if boundary, standard and method match. Traced figures are biased low.
Where should one household act first?	Consumer calculator, t CO ₂ e	Ranks private levers correctly. Contains a floor it cannot address and understates high-wealth emitters.
Does humanity exceed what the planet regenerates?	Ecological footprint, gha	Communicates overshoot in one figure. Dominated by its carbon term; unsuitable for ranking options.
What did an action change beyond the actor?	Handprint	Captures structural leverage. Never summed, never netted against a footprint.

If one practical step follows from all of this: compute a personal footprint once, to find the two or three levers that are actually large and to see the size of the floor underneath them. That floor is where the remaining majority sits, and it is not reachable by consumption decisions at all.

All charts were generated from the primary data files accompanying this document. Every source below was retrieved in July 2026. Research and drafting were carried out with assistance from Claude (Anthropic); every source cited was retrieved and checked against the primary document, and responsibility for the content rests with the author.

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